

[Turn over

- 1 (a) Fig. 1.1 shows the structure of a human heart.

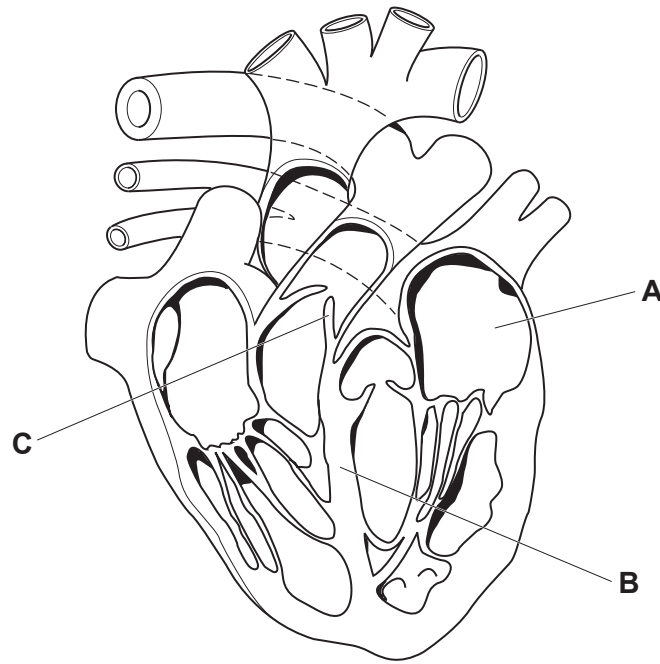


Fig. 1.1

- (i) Identify parts **A** and **B** shown in Fig. 1.1.

A

B

[2]

- (ii) Identify part **C** shown in Fig. 1.1.

..... [1]

- (iii) Describe the importance of part **C** to the flow of blood through the heart.

.....

..... [1]

(b) Complete Table 1.1 to show the function of different components of blood.

Table 1.1

components of blood	function
	produces antibodies
	transport hormones
platelets	

[3]

(c) Blood transports carbon dioxide around the body.

(i) Name the process in the body that produces carbon dioxide.

..... [1]

(ii) Carbon dioxide moves from the cells into the blood.

Complete the sentences to describe how carbon dioxide **moves out** of a cell into the blood.

The concentration of carbon dioxide is inside the cell than in the blood.

This causes carbon dioxide to move out of the cell by [2]

[Total: 10]

- 2 (a) A teacher uses the apparatus shown in Fig. 2.1.

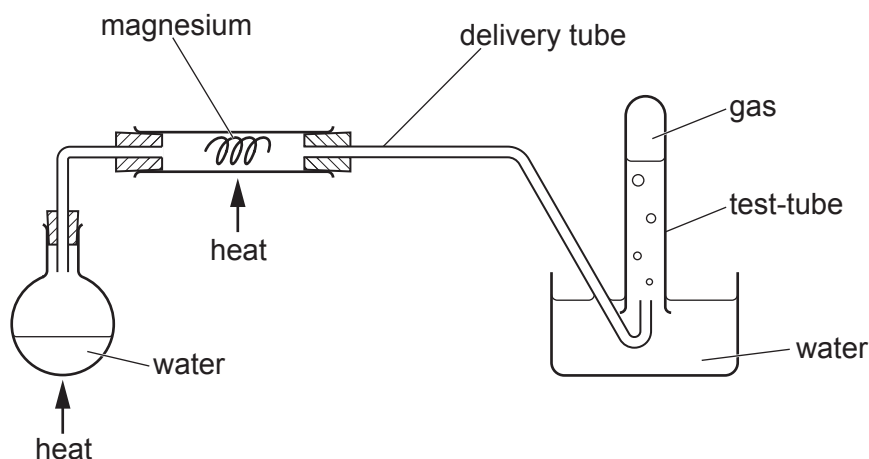


Fig. 2.1

The teacher heats the water to make steam.

The steam passes over heated magnesium.

The magnesium burns brightly. A white solid and a gas form.

The gas is collected in a test-tube. The teacher tests the gas using a lighted splint. It burns with a squeaky pop.

- (i) Identify **one** chemical change and **one** physical change in the teacher's experiment.

chemical change

physical change

[2]

- (ii) Give the chemical names for the white solid and the gas that form.

white solid

gas

[2]

- (b) An atom of sodium can be represented by the symbol shown.



Deduce the number of protons and the number of neutrons in this atom.

protons

neutrons

[2]

- (c) The electronic structure of a sodium atom is shown in Fig. 2.2.

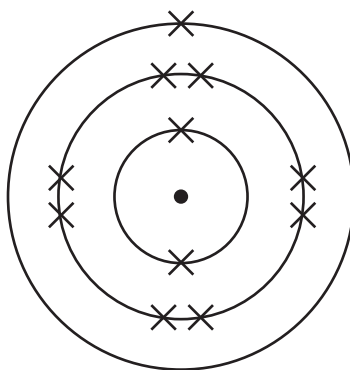


Fig. 2.2

Sodium reacts with chlorine to make sodium chloride.

Sodium chloride contains sodium ions, Na^+ , and chloride ions, Cl^- .

- (i) State the change to the electronic structure of sodium atoms when sodium reacts with chlorine.

..... [1]

- (ii) State which feature of the electronic structure of the sodium ion, Na^+ , makes the ion stable.

.....
..... [1]

[Total: 8]

- 3 Fig. 3.1 shows a rocket about to transport a large mirror into orbit around the Earth.

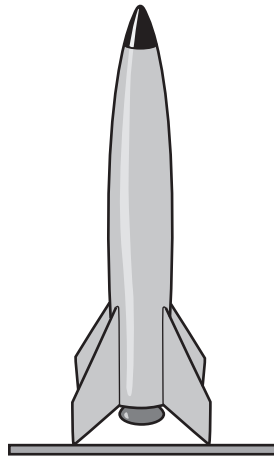


Fig. 3.1

- (a) The total mass of the rocket is 750 000 kg.
- (i) The Earth's gravitational field strength is 10 N/kg.

Calculate the weight of the rocket.

weight = N [1]

- (ii) When the rocket is launched, the force exerted on the rocket is 12 000 000 N vertically upwards.

Calculate the resultant force on the rocket.

State the direction of the resultant force.

resultant force = N

direction is [2]

- (iii) Describe the motion of the rocket as it leaves the Earth.

..... [1]

- (b) The rocket is powered by a fuel. The fuel is a store of chemical potential energy.

As the rocket moves upwards, large flames can be seen coming out of the back of the rocket.

The ground crew wear ear protection for their hearing as the rocket rises off the ground.

Use this information to identify **three** forms of energy resulting from the launch of the rocket.

1. energy
 2. energy
 3. energy
- [3]

- (c) In space, the rocket places a large mirror in orbit so that it reflects sunlight down to a solar panel on Earth.

Fig. 3.2 shows how the mirror is placed to reflect sunlight to a solar panel on Earth at night.

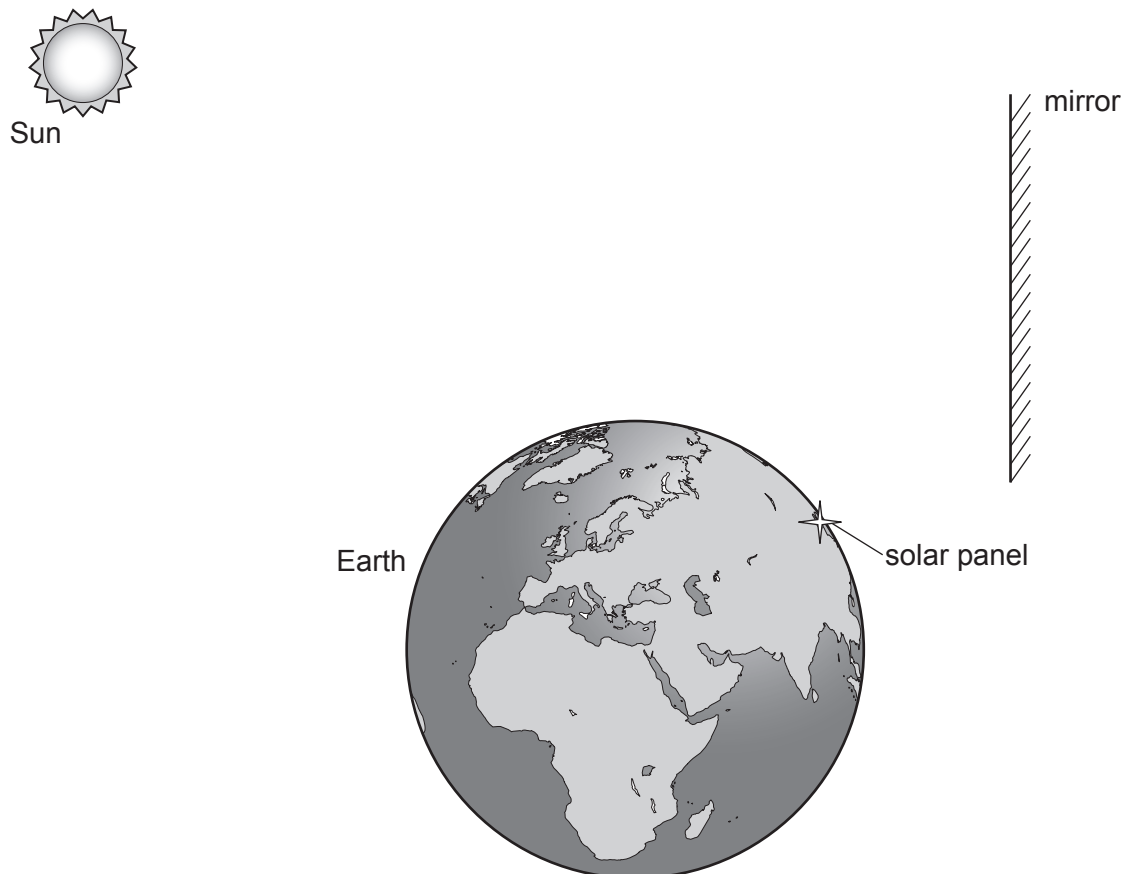


Fig. 3.2 (not to scale)

On Fig. 3.2, draw a ray to show how the mirror can reflect sunlight to the solar panel. [2]

[Total: 9]

- 4 (a) Fig. 4.1 is a graph showing the vitamin C content in 100 g of different foods.

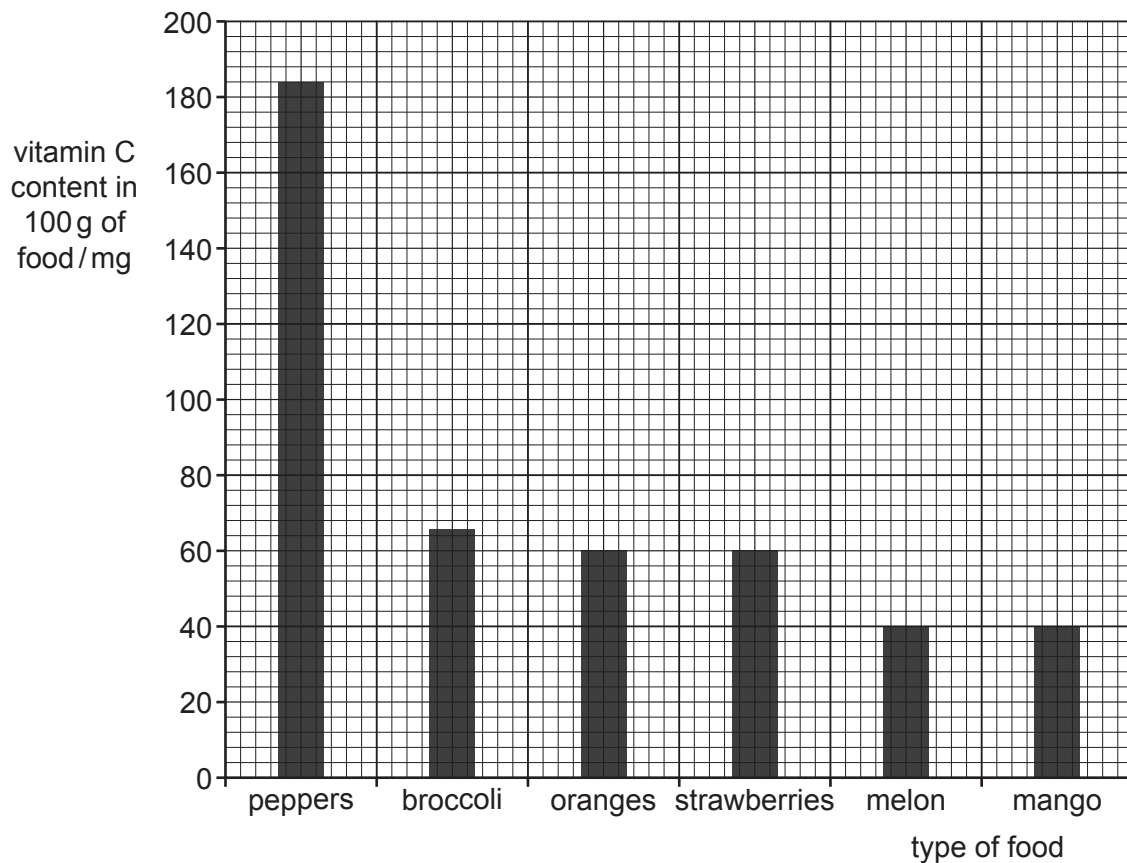


Fig. 4.1

- (i) Name **two** foods from Fig. 4.1 that contain exactly 60 mg of vitamin C in 100 g of the food.

..... and [1]

- (ii) Calculate the mass of mango needed to provide 80 mg of vitamin C.

mass =g [2]

(b) Fig. 4.2 shows the names and functions of some parts of the human alimentary canal.

Draw **one** line from the name of each part to its correct function.

name	function
anus	absorption
liver	ingestion
mouth	produces bile
small intestine	egestion

Fig. 4.2

[3]

(c) (i) Proteins and carbohydrates are made up of chemical elements.

List the chemical elements found in **both** proteins and carbohydrates.

..... [1]

- (ii) Proteins are digested by the enzyme pepsin.

Pepsin works best in the stomach.

The pH of the stomach is pH2.

Complete Fig. 4.3 by sketching a curve to show how the activity of the enzyme pepsin changes with pH.

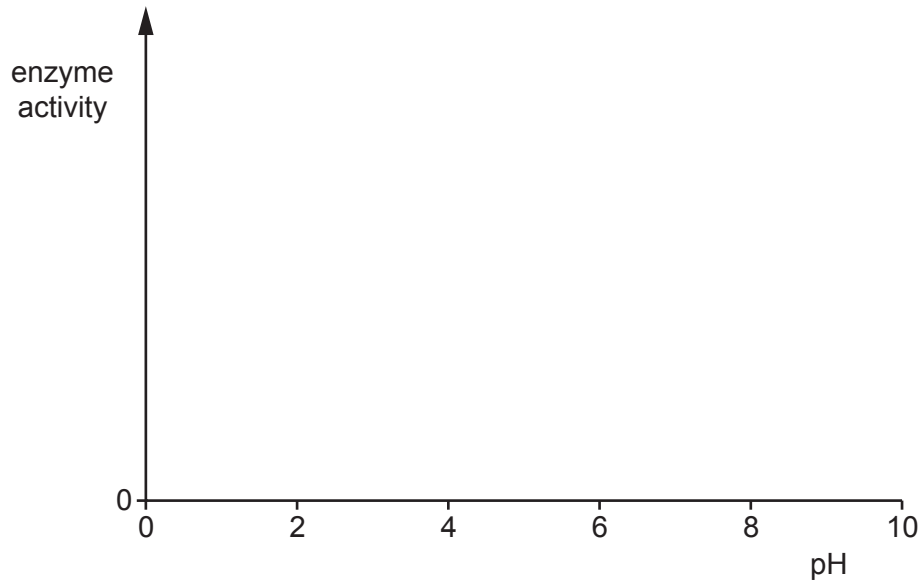


Fig. 4.3

[2]

- (iii) Proteins are made from small molecules that have reacted together.

Name these small molecules.

..... [1]

[Total: 10]

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- 5 (a) Petroleum is separated using the process shown in Fig. 5.1.

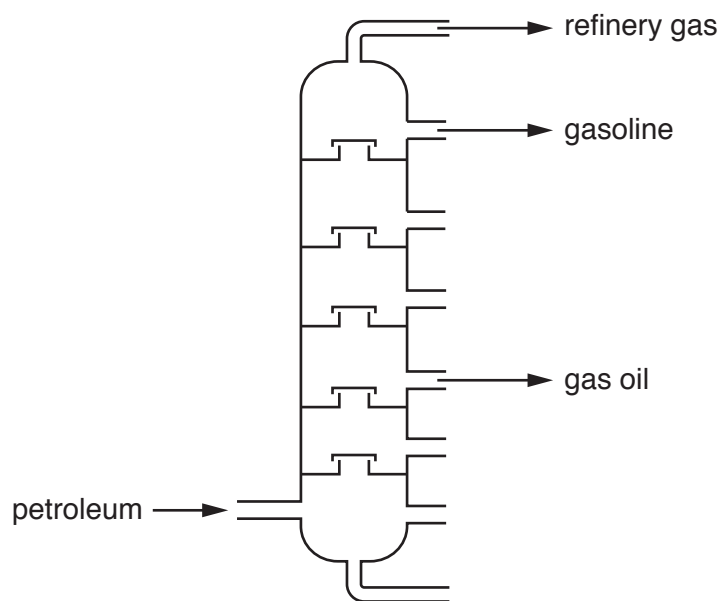


Fig. 5.1

- (i) Name the process shown in Fig. 5.1.

..... [1]

- (ii) Refinery gas and gas oil can be used as fuels.

State **one** use for each of these fuels.

refinery gas

gas oil

[2]

- (iii) Identify the greenhouse gas that forms during the complete combustion of gasoline.

..... [1]

- (b) Petroleum is a mixture of hydrocarbon molecules called alkanes.

- (i) Explain what is meant by *hydrocarbon*.

.....

..... [2]

- (ii) State the name of a process used to produce alkenes.

..... [1]

- (iii) Describe the difference between the bonding in alkanes and in alkenes.

.....

..... [1]

- (iv) Many ethene molecules combine to form poly(ethene).

State the type of addition reaction that occurs.

..... [1]

[Total: 9]

- 6 Fig. 6.1 shows a bucket of crushed ice used to cool drinks.

A thermometer is placed in the ice to check the temperature, and a glass bottle containing a drink is placed in the ice to cool.

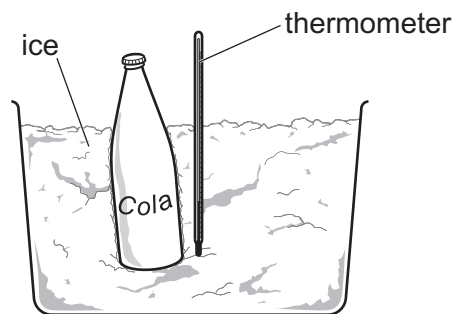


Fig. 6.1

The temperature of the ice when it is put in the bucket is -15°C .

The temperature of the drink before it is placed in the ice is 20°C .

- (a) Calculate the temperature difference between the ice and the drink at the start.

temperature difference = $^{\circ}\text{C}$ [1]

- (b) After 5 minutes the contents of the bucket are stirred and the temperature of the ice is taken again. The thermometer reading is -10°C .

State the names of **two** ways in which thermal energy has been transferred from the drink inside the bottle to the ice.

1.

2.

[2]

- (c) Fig. 6.2 shows a graph of how the temperature of the ice in the bucket changes over 30 minutes as the drink is cooled.

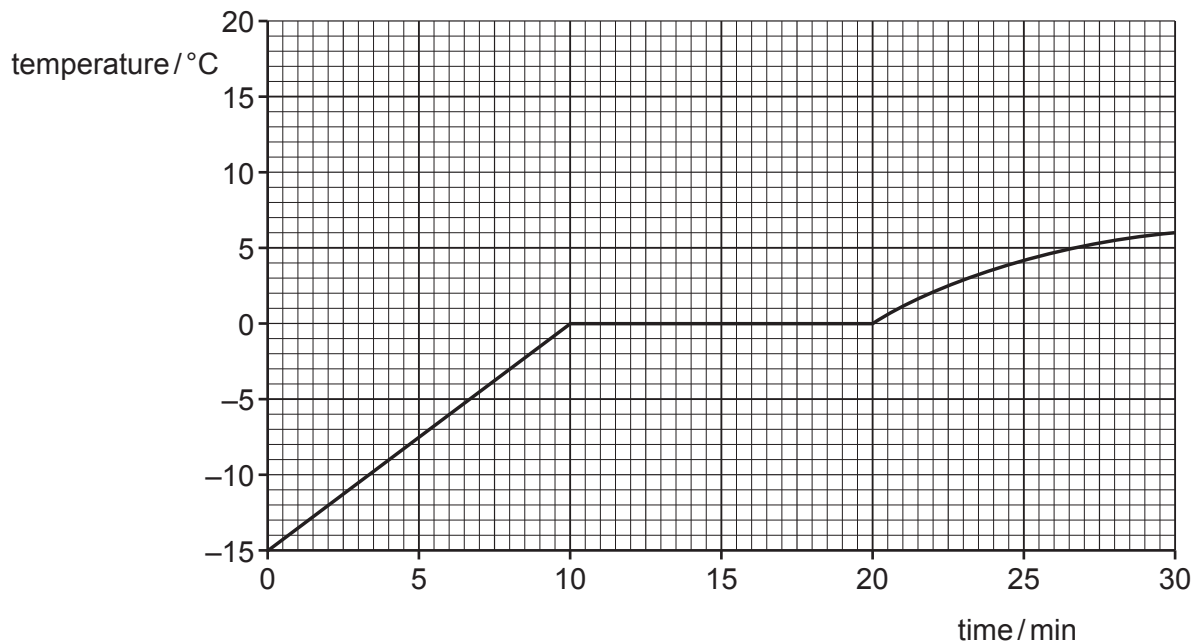


Fig. 6.2

- (i) State what is happening to the ice in the bucket between 10 and 20 minutes.
 [1]
- (ii) On Fig. 6.2, sketch a graph to show how the temperature of the drink in the bottle changes over the 30 minutes it is cooling in the ice bucket. [2]
- (d) At the start, the total mass of the bucket, ice and bottle of drink is 2.25 kg.
- (i) Predict the total mass of the bucket, ice and bottle of drink after 30 minutes.
 Give a reason for your answer.
 total mass = kg
 reason
 [1]
- (ii) The ice placed in the bucket is made from a block of mass 400 g.
 The volume of the block of ice is 436 cm^3 .
 Calculate the density of the ice.

density = g/cm^3 [2]

[Total: 9]

[Turn over]

7 (a) Fig. 7.1 shows a food web.

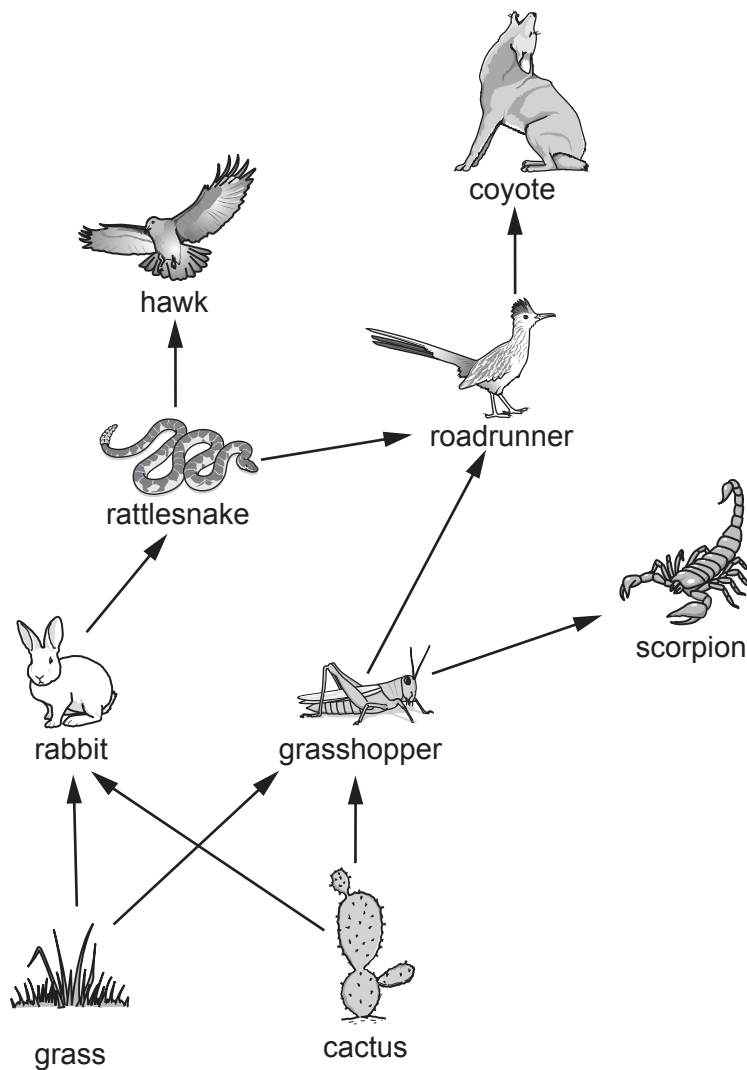


Fig. 7.1

(i) The food web is **not** complete.

Scorpions are eaten by roadrunners.

Use this information to complete the food web in Fig. 7.1 by drawing **one** arrow. [1]

(ii) Identify **one** tertiary consumer from the food web in Fig. 7.1.

..... [1]

- (iii) One year the number of rattlesnakes decreases.

This leads to a decrease in the number of grasshoppers.

Explain why using the information in Fig. 7.1.

.....

.....

.....

..... [2]

- (b) Fig. 7.2 is a diagram of a cactus growing in a desert.

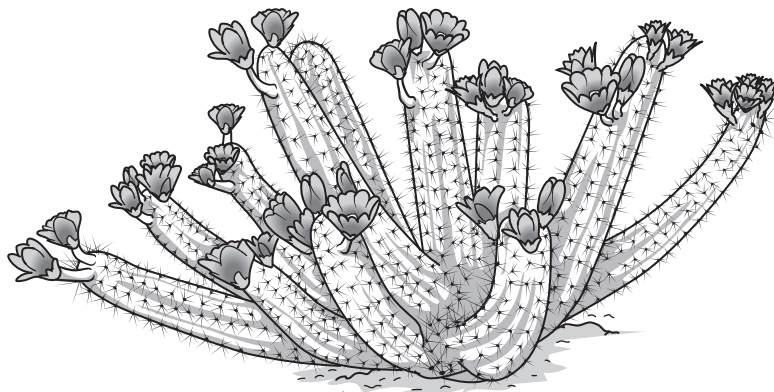


Fig. 7.2

The cactus is a flowering plant.

A bird feeds on the nectar produced by the flowers. As it feeds, pollen sticks to the bird and can be moved from one flower to another.

Complete the sentences using words from the list.

Each word may be used once, more than once or not at all.

agent

anther

catalyst

ovule

petals

sepals

stigma

The cactus flower is pollinated when pollen is transferred to the

The bird is the of pollination.

The flower has nectar and coloured to attract the bird. [3]

[Total: 7]

- 8 Period 4 of the Periodic Table is shown in Fig. 8.1. It contains the elements from potassium, K, to krypton, Kr.

Group																	
I	II											III	IV	V	VI	VII	VIII
19 K potassium 39	20 Ca calcium 40	21 Sc scandium 45	22 Ti titanium 48	23 V vanadium 51	24 Cr chromium 52	25 Mn manganese 55	26 Fe iron 56	27 Co cobalt 59	28 Ni nickel 59	29 Cu copper 64	30 Zn zinc 65	31 Ga gallium 70	32 Ge germanium 73	33 As arsenic 75	34 Se selenium 79	35 Br bromine 80	36 Kr krypton 84

Key

atomic number
atomic symbol
name
relative atomic mass

Fig. 8.1

- (a) State the trend in the metallic character of the elements from potassium to krypton.

..... to [1]

- (b) Iron is in a collection of elements between calcium, Ca, and gallium, Ga, in period 4.

These elements have high densities and form coloured compounds.

- (i) State the name of this collection of elements.

..... [1]

- (ii) Describe **one** other property shown by these elements.

..... [1]

- (c) Chlorine is a gas in Group VII.

It is made by passing electricity through an aqueous salt.

- (i) Name the process which uses electricity to break down an aqueous salt.

..... [1]

- (ii) Suggest **one** aqueous salt that is used to make chlorine in this process.

..... [1]

- (iii) Describe a chemical test for chlorine and the positive result for this test.

test

result

[2]

- (iv) State the name of the collection of elements in Group VII.

..... [1]

(d) Krypton is a gas in Group VIII (Group 0) in the Periodic Table.

Complete the sentence about this group.

The Group VIII (Group 0) gases are known as the gases. [1]

[Total: 9]

9 Fig. 9.1 shows a mobile (cell) phone and a loudspeaker.

The mobile phone transmits a signal to the loudspeaker.

The loudspeaker converts the signal into sound.

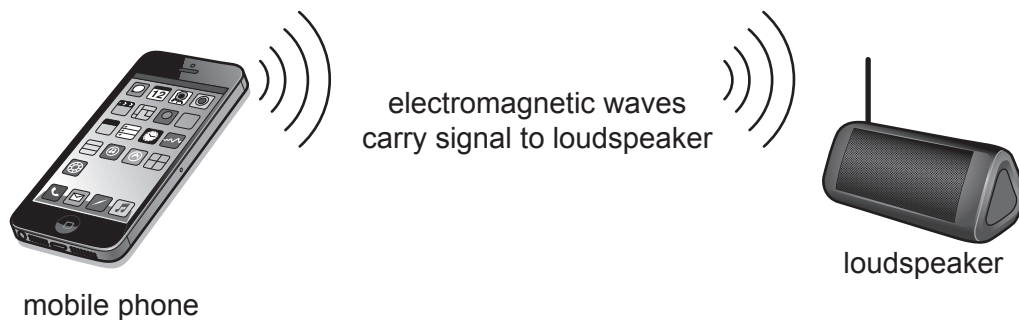


Fig. 9.1

(a) Electromagnetic waves of frequency 2.4×10^9 Hz are used to carry the signal from the mobile phone to the loudspeaker.

(i) Fig. 9.2 shows the approximate ranges of frequency for each type of electromagnetic wave.

more than 10^{19} Hz	10^{19} to 10^{17} Hz	10^{17} to 10^{15} Hz	10^{15} to 10^{14} Hz	10^{14} to 10^{11} Hz	10^{11} to 10^9 Hz	less than 10^9 Hz
gamma radiation	X-rays	ultraviolet	visible light	infrared	microwaves	radio waves

Fig. 9.2

Use the information in Fig. 9.2 to identify the type of electromagnetic wave used to carry the signal to the loudspeaker.

..... [1]

(ii) The loudspeaker can be switched on and off using a remote controller.

State the type of electromagnetic wave transmitted by a remote controller.

..... [1]

- (b) The loudspeaker is operated by an alternating current (a.c.) supply.

A lamp is connected in parallel with the loudspeaker.

A switch is used to switch on both the lamp and the loudspeaker.

- (i) On Fig. 9.3, complete the circuit diagram to show how the loudspeaker, lamp and switch are connected.

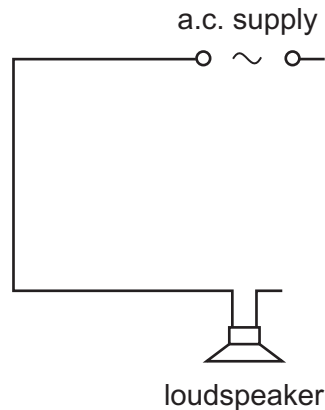


Fig. 9.3

[3]

- (ii) The loudspeaker requires a current of 0.80 A.

The power supply provides a voltage of 20 V across the loudspeaker.

Calculate the resistance of the loudspeaker.

Give the unit of your answer.

resistance = unit [3]

- (c) The alternating current supply has a frequency of 50 Hz.

When the speaker is switched on, but no signal is being received, sound waves of frequency 50 Hz are emitted from the loudspeaker.

Suggest whether a person near the loudspeaker will hear these sound waves.

Give a reason for your answer.

.....

 [1]

[Total: 9]

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The Periodic Table of Elements

Group											
I	II										
		1 Hhydrogen1									
		Key atomic number atomic symbol name relative atomic mass									
3 Li lithium 7	4 Be beryllium 9										2 He helium 4
11 Na sodium 23	12 Mg magnesium 24										10 Ne neon 20
19 K potassium 39	20 Ca calcium 40										18 Ar argon 40
37 Rb rubidium 85	38 Sr strontium 88										36 Kr krypton 84
55 Cs caesium 133	56 Ba barium 137										54 Xe xenon 131
87 Fr francium —	88 Ra radium —										86 Rn radon —
											85 At astatine —
											84 Lv livermorium —
											116 Lv livermorium —
											114 Fl flerovium —
											112 Cn copernicium —
											110 Ds darmstadtium —
											109 Mt meitnerium —
											108 Hs hassium —
											107 Bh bohrium —
											106 Sg seaborgium —
											105 Db dubnium —
											104 Rf rutherfordium —
											103 Ra actinoids —
											89–103 Ra actinoids —
											88 Ra actinoids —
											86 Rn radon —
											85 At astatine —
											84 Lv livermorium —
											83 Bi bismuth 209
											82 Pb lead 207
											81 Tl thallium 204
											80 Br bromine 80
											53 I iodine 127
											52 Te tellurium 128
											51 Sb antimony 122
											50 Sn tin 119
											49 In indium 115
											48 Cd cadmium 112
											47 Ag silver 108
											46 Pd palladium 106
											45 Rh rhodium 103
											44 Ru ruthenium 101
											43 Tc technetium —
											42 Mo molybdenum 96
											41 Nb niobium 93
											40 Zr zirconium 91
											39 Y yttrium 89
											38 Sr strontium 88
											37 Rb rubidium 85
											36 Kr krypton 84
											35 Br bromine 80
											34 Se selenium 79
											33 As arsenic 75
											32 Ge germanium 73
											31 Ga gallium 70
											30 Zn zinc 65
											29 Cu copper 64
											28 Ni nickel 59
											27 Co cobalt 59
											26 Fe iron 56
											25 Mn manganese 55
											24 Cr chromium 52
											23 V vanadium 51
											22 Ti titanium 48
											21 Sc scandium 45
											20 Ca calcium 40
											19 K potassium 39
											18 Ar argon 40
											17 Cl chlorine 35.5
											16 S sulfur 32
											15 P phosphorus 31
											14 Si silicon 28
											13 Al aluminium 27
											12 C carbon 12
											11 B boron 11
											10 Ne neon 20
											9 F fluorine 19
											8 O oxygen 16
											7 N nitrogen 14
											6 C carbon 12
											5 B boron 11
											4 He helium 4

lanthanoids	57	La	lanthanum	139	58	Ce	cerium	140	59	Pr	praseodymium	141	60	Nd	neodymium	144	61	Pm	promethium	—	62	Sm	samarium	150	63	Eu	europtium	152	64	Gd	gadolinium	157	65	Tb	terbium	159	66	Dy	dysprosium	163	67	Ho	holmium	165	68	Er	erbium	167	69	Tm	thulium	169	70	Yb	yterbium	173	71	Lu	lutetium	175
	89	Ac	actinium		90	Th	thorium	232	91	Pa	protactinium	231	92	U	uranium	238	93	Np	neptunium		94	Pu	plutonium		95	Am	americium		96	Cm	curium		97	Bk	berkelium		98	Cf	californium		99	Es	einsteinium		100	Fm	fermium		101	Md	mendelevium		102	No	nobelium		103	Lr	lawrencium	
	actinoids																																																											

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).